Progress Report

Distances in Adjectives: Comparing Lexical Profiles of Telecom Advertising

Marta Trales

Abstract—We analyze lexical distances between adjectives consumers associate with ads from Orange and Bouygues Telecom, using 800 synthetic interviews generated by Mistral-7B. Adjectives are extracted via spaCy (French) and compared using Jaccard similarity, Jensen–Shannon divergence, cosine similarity, and Shannon entropy. Results show low brand-level overlap (mean Jaccard ≈ 0.16), distinct per-brand vocabularies, and a human-agent preference agreement of 27.6%.

1. PROJECT OVERVIEW

This step investigates the *lexical distance* between adjectives that consumers associate with advertisements from two major French telecom operators: **Orange** and **Bouygues Telecom**. The core hypothesis is that the two brands elicit semantically distinct emotional and evaluative vocabularies, and that this divergence can be measured and quantified using standard NLP distance metrics.

The analysis is grounded in a dataset of 800 synthetic consumer interviews, generated by `Mistral-7B-Instruct` and collected in November 2025. Each interview simulates a consumer watching both ads and responding to structured questions on spontaneous reactions, memorability, attractiveness, emotional resonance, brand image, and purchase intent.

2. DATA DESCRIPTION

2.1. Datasets

Three data files are loaded at the start of the notebook:

- `resultats\_extraits\_adj\_g3.csv` — Per-panelist adjective lists for Orange and Bouygues, extracted directly from short-form preference responses.
- `preference\_resultats\_mistral\_7b\_800.csv` — Agent-predicted brand preference labels and confidence scores (0–1) per panelist.
- `interviews.csv` — Full interview panel with 800 rows and 48 columns, including demographics and 24 open-ended response fields.

2.2. Panelist Demographics

The synthetic panel spans a range of socio-demographic profiles, including age (from 23 to 68 in the preview), gender, CSP (socio-professional category), education level, income, and urban/rural classification. This diversity is intentional to simulate a representative French consumer sample.

3. NLP PIPELINE

3.1. Adjective Extraction

Full-text adjective extraction is performed using the `fr\_core\_news\_lg` model from spaCy. For each of the 24 response columns (8 Bouygues, 8 Orange, 6 comparison), adjectives are identified by POS tagging and lemmatized to lowercase. The results are aggregated per panelist into two lists: `adj\_orange\_full` and `adj\_bouygues\_full`.

This progress report documents the exploratory data analysis and NLP pipeline applied to a corpus of 800 consumer interviews comparing advertising spots from Orange and Bouygues Telecom. Data collected November 2025.

3.2. Corpus-Level Token Distributions

Token frequencies are aggregated across all panelists using Python's `Counter`. The resulting probability distributions are used for information-theoretic distance metrics. The top 20 adjectives per brand are shown in the following tables.

Table 1. Top 10 Adjectives — Orange (lemmatized, all interviews)

Adjective	Count
fiable	484
simple	329
positif	242
humaine	170
classique	125
bonne	123
claire	108
chaleureux	107
original	102
positive	93

Note: Counts reflect raw token frequency summed across all 800 panelists and all 24 interview questions.

Table 2. Top 10 Adjectives — Bouygues Telecom (lemmatized, all interviews)

Adjective	Count
bizarre	179
original	169
fiable	154
positif	140
créatif(ve)	121
originalité	105
originale	101
créatif	101
fort	88
différent	84

Note: Counts reflect raw token frequency summed across all 800 panelists and all 24 interview questions.

4. DISTANCE METRICS

4.1. Per-Panelist Jaccard Similarity

For each panelist, the Jaccard similarity between their Orange and Bouygues adjective sets is computed as:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{1}$$

Across 800 panelists, the distribution of per-panelist Jaccard scores is summarized in Table 3. The mean score of 0.159 indicates very low adjective overlap, suggesting that consumers use largely distinct vocabularies for each brand.

Table 3. Per-Panelist Jaccard Similarity Statistics

Metric	Jaccard	n overlap	n Orange-only
Mean	0.159	1.470	4.418
Std	0.154	1.369	1.985
Min	0.000	0.000	0.000
Median	0.125	1.000	4.000
Max	0.889	8.000	10.000

n overlap: shared adjectives per panelist; n Orange-only: exclusive to Orange.

4.2. Corpus-Level Divergence

At the aggregate level, the following metrics are computed over the normalized token distributions:

- **Cosine similarity:** 0.6394 — moderate overlap in high-frequency terms (e.g., *fiable*, *simple*).
- **Jensen–Shannon divergence:** 0.5411 — indicates substantial distributional divergence.
- **KL divergence (Orange → Bouygues):** ∞ — Orange uses adjectives with zero probability in Bouygues (and vice versa), confirming non-overlapping vocabulary regions.
- **Shannon entropy (Orange):** 6.81 bits; **(Bouygues):** 7.38 bits — Bouygues elicits a more diverse and unpredictable vocabulary.

5. VOCABULARY ANALYSIS

5.1. Shared and Exclusive Adjectives

Out of all unique lemmatized adjectives used in the corpus:

- **303** adjectives are shared between both brands.
- **215** are exclusive to Orange.
- **259** are exclusive to Bouygues.

Top Orange-only adjectives include: *connu, générale, convenu, diffus, tranquillité, authenticité, sécurité, simplicité*. These reflect a **trust-and-reliability** semantic cluster consistent with Orange's brand messaging ("la fibre la plus fiable").

Top Bouygues-only adjectives include: *audace, parodique, visuelle, étrangeté, policier, intellectuel, mini-film, magique, cinéma, marrant*. These reflect a **humor-and-creativity** semantic cluster aligned with Bouygues' more narrative, comedic ad concept.

5.2. Adjective Overlap by Preference

Grouping panelists by stated preference (human) reveals a striking pattern (see Table 4): panelists who prefer Orange show *higher* Jaccard similarity and more overlap adjectives, while Bouygues-preferring panelists use more distinct, non-overlapping vocabulary per brand.

Table 4. Adjective Overlap by Preference Group

Prefers Bouygues	Jaccard	n overlap	n Orange-only	n Bouygues-only
False (Orange)	0.286	2.000	2.500	1.000
True (Bouygues)	0.159	1.469	4.422	3.862

Values are per-panelist means within each preference group.

6. HUMAN VS. AGENT AGREEMENT

A key validation step compares the brand preference inferred by the Mistral-7B agent (from the `preferences` dataset, `brand` column) with the human-annotated preference from the interview text.

$$\text{Agreement} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{y}_i = y_i] \tag{2}$$

The resulting overall agreement is $\approx 27.6\%$, which is notably low. This suggests either a systematic misalignment between how the agent summarizes preferences and how the raw text expresses them, or that the preference extraction task is inherently ambiguous and requires further prompt refinement.

Note

The low agreement rate (27.6%) does not necessarily indicate a failure. Many panelists express nuanced, context-dependent preferences (e.g., preferring Bouygues creatively but Orange commercially). Future work should refine the preference extraction schema to distinguish between aesthetic and purchase-intent preference.

7. CURRENT STATUS AND NEXT STEPS

7.1. Completed Work

- Data loading and merging of all three sources (*resultats*, *preferences*, *panel*).
- Corpus-level adjective frequency profiling for Orange and Bouygues.
- Per-panelist Jaccard similarity computation and descriptive statistics.
- Full adjective extraction from all 24 interview response fields via spaCy (`fr\_core\_news\_lg`).
- Corpus-level distributional metrics: cosine similarity, Jensen–Shannon divergence, KL divergence, Shannon entropy.
- Vocabulary split analysis (shared vs. brand-exclusive adjectives).
- Human-vs-agent agreement rate calculation.

7.2. Planned Work

- **Clustering:** Apply k-means or hierarchical clustering over panelist adjective vectors to identify latent perception profiles.
- **Dimensional reduction:** Visualize adjective distributions using UMAP or PCA on TF-IDF or embedding representations.
- **Regression analysis:** Model Jaccard similarity or brand preference as a function of demographic variables (age, CSP, income, location).
- **Agreement refinement:** Revisit agent-generated preference labels; investigate confidence score thresholds and prompt strategies.
- **Semantic embedding distances:** Replace bag-of-words Jaccard with sentence-transformer-based semantic similarity between full adjective lists.